https://kubernetes.io/docs/concepts/workloads/pods/probes/

What are Kubernetes Probes?

Kubernetes uses probes to check the health of containers.

Kubernetes lets you define *probes* to continuously monitor the health of containers in a Pod. A probe is a diagnostic performed periodically by the kubelet on a container. To perform a diagnostic, the kubelet either executes code within the container or makes a network request.

Based on the probe results, Kubernetes can restart unhealthy containers or stop sending traffic to containers that are not ready.

Probe	Purpose	Failure Action
Liveness	Is the application still alive?	Kubernetes restarts container
Readiness	Can the application receive traffic?	Pod removed from Service endpoints
Startup	Has the slow-starting application started?	Liveness/readiness are held back

Simple way to remember

Startup → Has my application started?

Readiness → Can I send traffic to it?

Liveness → Is my application still healthy?